

KENDRIYA VIDYALAYA SANGATHAN, RAIPUR REGION
FIRST PRE-BOARD EXAM (2025-26)
SUBJECT - SCIENCE (086)

CLASS: X

MAX MARKS: 80

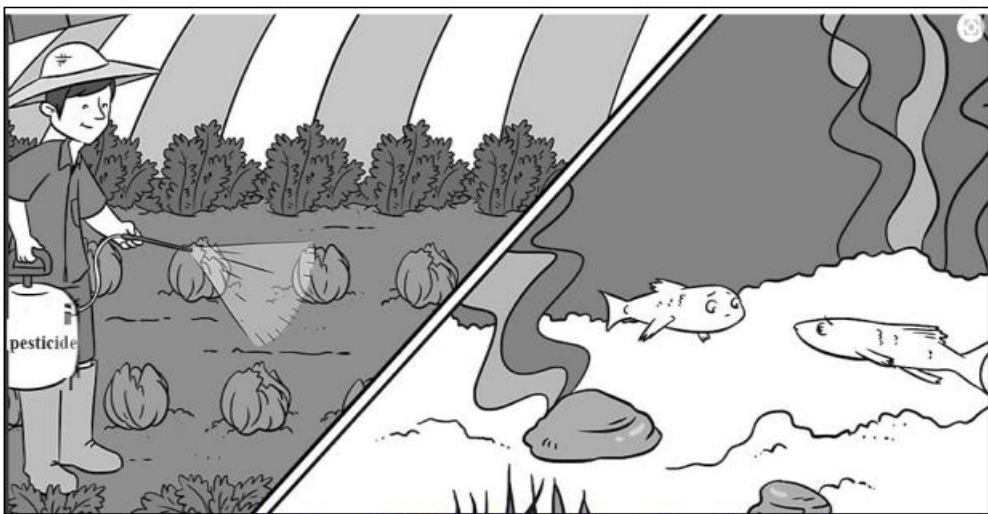
TIME: 03 HOURS

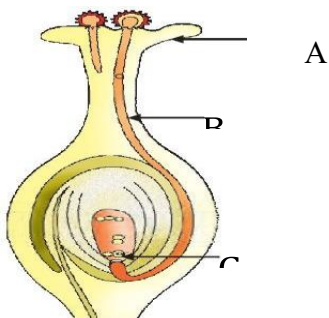
General Instructions:

1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
3. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

NO.	SECTION A	MAR KS
1.	Secretion of less saliva in mouth will affect the conversion of : A. proteins into amino acids B. fats into fatty acids and glycerol C. starch into simple sugars D. sugars into alcohol	1
2.	Select the correct option from the following statements about the functioning of the human heart. A. Right atrium receives deoxygenated blood from different parts of the body and sends it to pulmonary veins. B. Left atrium sends oxygenated blood to right ventricle which pumps it to different parts of the body. C. Right atrium receives deoxygenated blood from the body and sends it to the right ventricle. D. Left atrium receives oxygenated blood from the pulmonary arteries and sends it to the left ventricle.	1
3.	Observe the given figures A and B. When <i>chhui-mui</i> (sensitive) plant is touched, its leaves fold. This is due to :   Figure AFigure B A. Hormonal effect B. Thermal effect C. Change in amount of water in cells D. Electromagnetic effect	1

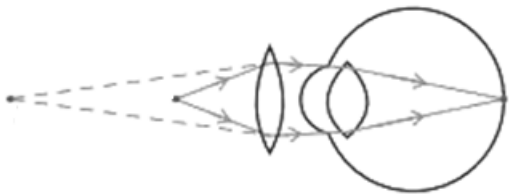
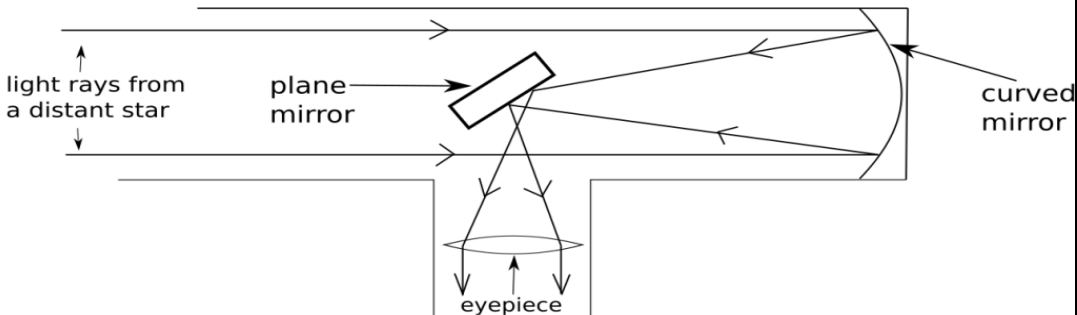
4.	The blood glucose level in a patient was very high. It may be due to inadequate secretion of: A. growth hormone from pituitary gland B. oestrogen from ovary C. insulin from pituitary gland D. insulin from pancreas	1								
5.	Identify from the following a group containing all non-biodegradable substances. A. Leather, Glass, Plastic B. Cotton, Wood, Nylon C. DDT, Polyester, Glass D. Leather, Silk, Wool	1								
6.	Which are the correct statements related to ozone? (i) Ozone layer helps in increasing the UV radiations reaching earth. (ii) Ozone is a deadly poison. (iii) Ozone layer shields the earth from UV radiations. (iv) Ozone layer prevents UV rays which cause skin cancer. (v) Ozone is formed with the help of Chlorofluorocarbons. A. (i), (ii), (iii) B. (ii), (iii), (iv) C. (iii), (iv), (v) D. (i), (iv), (v)	1								
7	If pea plants with round and green seeds (RRyy) are crossed with pea plants having wrinkled and yellow seeds (rrYY), the seeds developed by the plants of F ₁ generation will be: A. 50% round and green B. 75% wrinkled and green C. 100% round and yellow D. 75% wrinkled and yellow	1								
	The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true.									
8.	Assertion(A) : The offspring produced by sexual reproduction is likely to adjust better in environmental fluctuation. Reason (R) : During the fusion of gametes there is mixing of genetic material from two parents.	1								
9.	Assertion(A) : The sex of a child is determined by the mother. Reason (R) : Humans have two types of sex chromosomes: XX and XY.	1								
10.	(I)The transport system in plants is relatively slower than in animals. Give reasons. (II)State the role of phloem in the transport of materials in plants. OR Besides minimizing the loss of blood, why is it essential to plug any leak in a blood vessel? Name the component of blood which helps in this process and state how this component performs this Function.	2								
11.	Differentiate between alveoli and nephron on the basis of the following points: <table><tr><td>S. No.</td><td>Feature</td><td>Alveoli</td><td>Nephron</td></tr><tr><td>1</td><td>Structure and location</td><td></td><td></td></tr></table>	S. No.	Feature	Alveoli	Nephron	1	Structure and location			2
S. No.	Feature	Alveoli	Nephron							
1	Structure and location									

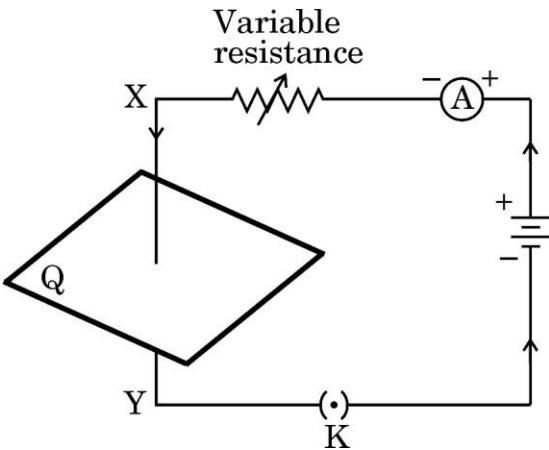
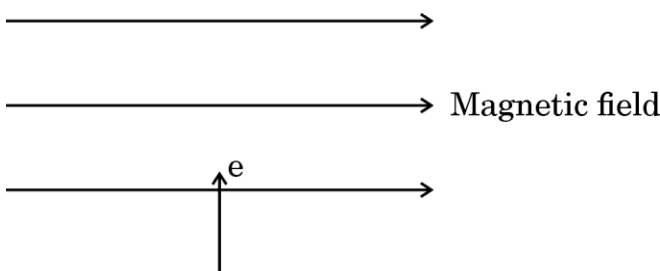
	2	Function			
12.	The cartoon below addresses a growing concern:  What impact will the process shown in the image have on Humans if they occupy the last trophic level? Explain.				2
13	(I) Define hormone. (II)“Hormones should be secreted in precise quantities. We have a feedback mechanism through which this is done.” With the help of an example justify the statement.				1+2=3
14	A pure pea plant bearing terminal flowers was cross-pollinated with a pure plant having axillary flowers. In F₁ generation, plants with axial flowers only were obtained. F₁ generation plants are self-pollinated and F₂ generation is obtained. (I)Work out the pattern of inheritance in this case. (II))What will be the ratio of plants obtained in F₂ generation ?				2+1=3
15	The maintenance functions of all living organisms must go on even when they are not doing anything particular. Even when we are just sitting in a class or even asleep, this maintenance job has to go on. These maintenance processes require energy to prevent damage and break-down of cells and tissues, which is obtained by the individual organism from the food prepared by the autotrophs, called producers. (I)Name and define the process by which green plants prepare food. (II)Write chemical equation involved in the above process. (III)State in proper sequence the events that occur in synthesis of food by desert plants.				1+1+2

16	Differentiate between self-pollination and cross-pollination. Identify A, B and C in the diagram given below and write one function of each. 	2+3
SECTION B		
17.	Identify 'p', 'q' and 'r' in the following balanced reaction $p \text{ Pb}(\text{NO}_3)_2(\text{s}) \xrightarrow{\text{Heat}} q \text{ PbO}(\text{s}) + r \text{ NO}_2(\text{g}) + \text{O}_2(\text{g})$ A. 2,2,4 B. 2,4,2 C. 2,4,4 D. 4,2,2	1
18	4.Consider the following reactions : (i)Dilute hydrochloric acid reacts with sodium hydroxide. (ii)Magnesium oxide reacts with dilute hydrochloric acid. (iii)Carbon dioxide reacts with sodium hydroxide. It is found that in each case : (A)Salt and water is formed. (B)Neutral salts are formed. (C)Hydrogen gas is formed. (D)Acidic salts are formed.	1
19	The warning sign shown in the given figure must invariably be displayed/pasted on the containers which contain hydroxide of :  (A)Aluminium (B) Calcium (C) Sodium (D) Magnesium	1
20	Aluminium powder is used in thermit welding because : A. Its reaction with iron is highly exothermic. B. When it is heated with iron (III) oxide, molten iron is obtained.	1

	C. When it is heated with iron (III) oxide, molten aluminium oxide is obtained to join railway tracks. D. Its melting point is low as compared to iron and a molten alloy of iron and aluminium is formed on heating which is used to join railway tracks.	
21	When 50g of lead powder is added to 300 ml of blue copper sulphate solution, after a few hours, the solution becomes colourless. This is an example of A. Combination reaction B. Decomposition reaction C. Displacement reaction D. Double displacement reaction	1
22	The electronic configuration of three elements X, Y and Z are X- 2, 8, 7; Y- 2, 8, 2; and Z - 2, 8 A. Y and Z are metals B. Y and X are non-metals C. X is a non-metal and Y is a metal D. Y is a non-metal and Z is a metal	1
23	Which of the following is an endothermic reaction? A. Burning of candle. B. Cooking of food. C. Decomposition of Vegetable matter. D. Reaction of Sodium with air	1
24	Assertion(A) : Carbon compounds can form chain, branched and ring structures. Reason (R) : Carbon exhibits the property of catenation. Answer the question by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true.	1
25	(a) In common practice silver is recovered from silver nitrate solution by the use of copper metal. Name the type of reaction that takes place in this process and give the chemical equation of the reaction involved. (b) Name the method used for refining silver.	2
26	A. Anirudh took two metal oxides; aluminium oxide and magnesium oxide as shown in the pictures given below. But he forgot to label them. How will you guide/ help Anirudh to identify the oxides and label them?   B. In an activity Aishu was given two substances; Copper Sulphide (Cu₂S) and Copper Oxide (Cu₂O) to obtain copper from these compounds. She was able to extract Copper successfully. Illustrate with the help of chemical	1+2

	equations how Aishu might have completed the activity	
27	(I) Name two metals which react violently with cold water. List any three observations which a student notes when these metal are dropped in a beaker containing water. (II) Write a test to identify the gas evolved (if any) during the reaction of these metals with water.	2+1
28	Corrosion is the phenomenon of deterioration of surface of metal in presence of air and moisture. It is a natural process and in the presence of a moist atmosphere, chemically active metals get corroded. This is an oxidation reaction. Rusting is the process where iron corrodes due to exposure to the atmosphere. The main circumstance of corrosion occurs with iron because it is a structural material in construction, bridges, buildings, rail transport, ships, etc. Aluminium is also an important structural metal, but even aluminium undergoes oxidation reactions. However, aluminium doesn't corrode or oxidize as rapidly as its reactivity suggests. Copper (Cu) corrodes and forms a basic green carbonate. (I) What is rusting? (II) Corrosion is (a) a redox reaction (b) a reduction reaction (c) a displacement reaction (d) an oxidation reaction (III) Which two metals do not corrode easily? (IV) Write the chemical name of the compound formed on corrosion of silver.	4
29	(i) Give reason why carbon forms compounds mainly by covalent bonding. (ii) Why do covalent compounds have low melting and boiling points. (iii) Give reason for the following : I. Covalent compounds are bad conductors of electricity. II. Carbon shows catenation III. Why graphite is good conductor of electricity	5
	SECTION -C	
30	Three resistors of 1 Ω, 2 Ω and 3 Ω are connected in parallel. The combined resistance of the three resistors should be (a) greater than 3 Ω (b) less than 1 Ω (c) equal to 2 Ω (d) between 1 Ω and 3 Ω	1
31	The part of the human eye which can modify the curvature of the eye lens to some extent is : A. Pupil B. Cornea C. Ciliary muscles D. Aqueous humour	1
32	Assertion (A) : Heater wire must have high resistance and high melting point. Reason (R) : If resistance is high, the electric conductivity will be less. Answer the question by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true.	1

33	A ray of light after refraction from a convex lens emerges parallel to its principal axis. I. Draw a labelled ray diagram to show it. II. In this case, the incident ray before refraction from the lens passes through a III. point on its principal axis. Name the point.	2
34	How and why is an electric fuse used in an electric circuit? Briefly describe its function.	2
35	A. State the relationship between the resistance R of a wire to its length l and cross-sectional area A. Use the mathematical symbols to arrive at the final formula. B. Using the formula define the resistivity of a material.	3
36	Draw a ray diagram to show the path of the reflected ray in each of the following cases. A ray of light incident on a convex mirror : (a) strikes at its pole making an angle θ from the principal axis. (b) is directed towards its principle focus. (c) is parallel to its principal axis.	3
37	 The above image shows a corrective measure for a particular defect of vision. (i) Identify the defect of vision and state what kind of lens is used to correct this deficiency. (ii) Draw and label a ray diagram that shows the defect of vision in the above case before correction. OR (i) What is dispersion of light? (ii) Explain the condition under which dispersion happens? (iii) Give one reason that causes presbyopia.	3
38	 The above image is that of a reflecting telescope. Reflecting telescopes revolutionised our ways of looking into the sky. They employ mirrors to gather and focus light, rather than relying solely on lenses as in their refracting counterparts. These telescopes utilise precisely shaped and polished mirrors to capture incoming light and reflect it to a focal point, where it forms an image	4

	for observation. A. What kind of image of the star is seen by the observer at the eyepiece? B. What kind of mirror is used in this reflecting telescope? C. Explain with reason what kind of optical device (type of lens or mirror) that is used at the eyepiece.	
39	The given figure shows the current passing through the straight conductor XY.  I. Copy the diagram and draw the magnetic field lines when current flows from conductor X to Y. II. Name and state the rule used in determining the direction of the magnetic field lines in the situation given above. III. State Fleming's left hand rule. Using this rule, determine the direction of force applied on an electron entering a uniform magnetic field as shown in the figure.  OR I. Define the term solenoid. Draw the pattern of the magnetic field lines in and around a current carrying straight solenoid. Mark on the pattern the (i) direction of current, (ii) direction of field lines near the ends of the solenoid, and (iii) region where the magnetic field is uniform. II. How would you make an electromagnet using a current carrying solenoid ?	5